Electric-Vehicle Supply Equipment Network Optimization in the Kitchener-Waterloo-Cambridge CMA

by

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- OpenChargeMap API for comprehensive charging infrastructure data
- Statistics Canada for detailed demographic and population data
- Ontario's FSA-level EV Database for electric vehicle ownership insights
- GRT for their open data on transit network infrastructure
- OpenStreetMap and OSMnx [1] for geospatial analysis capabilities

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List of Abbreviations

AC Alternating Current 1, 2

DCFC Direct Current Fast Charging 2

EV electric vehicles 1, 2, 4, 5, 11, 12, 22

FSA Forward Sortation Area ii, 4, 5

GRT Grand River Transit ii

IESO Independent Eletricity System Operator 4, 5

KWC-CMA Kitchener-Waterloo-Cambridge Census Metropolitan Area 1, 10, 12

MIP Mixed-Integer Program 2, 6, 10, 12

UTM Universal Transverse Mercator 23

Chapter 1

Introduction

The transition to EV (electric vehicles) represents a fundamental shift in transportation infrastructure requirements across Canada. Recent data from the Canada Energy Regulator (2024) indicates unprecedented growth in EV adoption, with sales reaching 12.8% of new vehicle registrations in 2023, marking a significant milestone in sustainable transportation [2]. This rapid adoption aligns with federal initiatives supporting the Paris Agreement and stems from policy frameworks such as the Canadian Net Zero Emissions Accountability Act, which mandates complete transition to zero-emission vehicle sales by 2035 [3].

The KWC-CMA (Kitchener-Waterloo-Cambridge Census Metropolitan Area) exemplifies both the opportunities and challenges of this transition. Despite strong EV adoption trends documented in Ontario Data Catalogue [4], our analysis reveals that current charging infrastructure reaches only 14.57% of the population for Level 2 charging and a mere 1.78% for Level 3 charging, creating a critical disparity between EV adoption and infrastructure availability. The Hydro One Kitchener-Waterloo-Cambridge-Guelph Regional Infrastructure Plan (2021) highlights the importance of coordinated infrastructure development to support increasing electrical demand from EV adoption [5] [6].

This study addresses these challenges through a comprehensive optimization framework to improve the KWC-CMA EV charging infrastructure. To contextualize this framework, it is essential to understand the technological landscape of EV charging equipment, which is typically classified into three distinct tiers. Level 1 chargers utilize standard 120V AC (Alternating Current) outlets, delivering approximately 1.4 to 1.9 kW of power. While highly accessible, L1 charging is characterized by prolonged charging durations, often requiring overnight sessions to meaningfully replenish a vehicle's range, making it primarily suited for residential applications. Level 2 charging represents a significant step up, op-

erating on 208V to 240V AC supplies and providing between 3.2 kW and 19.2 kW of power. Literature indicates that L2 infrastructure serves as the backbone of public and workplace charging networks, offering a pragmatic balance between installation costs and moderate charging speeds that can service vehicles over several hours. In contrast, Level 3, or DCFC (Direct Current Fast Charging), supplies high-voltage direct current directly to the battery, bypassing the onboard converter. Level 3 stations offer vastly superior power supply capacities, typically ranging from 50 kW to over 350 kW, enabling them to recharge an EV battery to 80% capacity in under 30 minutes. This rapid charging speed is critical for facilitating long-distance travel and alleviating range anxiety among drivers. We employ a Mixed-Integer Program to optimize the placement of new charging stations while considering strategic upgrades of existing Level 2 stations to Level 3 capabilities, directly addressing the varying power demands and duration constraints identified in recent literature. Our approach integrates multiple data streams to develop cost-effective solutions that maximize charging coverage. The insights gained from this study can inform policy development, guide infrastructure investments, and support the broader transition to sustainable urban mobility.

Chapter 2

Data Collection and Integration

The project integrates multiple authoritative Open Data sources and implements rigorous processing techniques to develop a robust framework for spatial analysis and infrastructure planning. See Figure 1 in Appendix for data pipeline visualization.

2.1 Geographic and Infrastructure Data

2.1.1 Boundary Data

The foundation of our analysis rests on comprehensive geographic data from the Region of Waterloo's Open Data Portal [7]. We employed the UTM Zone 17N Projection system to ensure precise spatial measurements throughout our analysis. This geographic framework proved essential for accurate coverage calculations and infrastructure planning, particularly in areas where service boundaries intersect with major transportation corridors.

2.1.2 Charging Infrastructure

Our infrastructure assessment through the OpenChargeMap API [8] revealed a network of 183 charging stations across the KWC-CMA region. The existing infrastructure comprises 167 Level 2 stations providing 424 charging ports, operating at the standard 19.2 kW per port capacity specified by HoneyBadger Charging (2023) [9]. Additionally, 16 Level 3 stations offer 66 fast-charging ports, each capable of delivering up to 150 kW as documented

by Future Energy (2022) [10]. The spatial distribution of these stations formed a crucial baseline for our optimization model.

2.1.3 Potential Locations

OpenStreetMap data, accessed through OSMnx [1], provided the basis for identifying potential charging station locations. Our analysis considered commercial centres, retail locations, public parking facilities, and community venues, with particular attention to grid infrastructure proximity as detailed in IESO planning documents [11]. The evaluation of each location incorporated land use compatibility, site accessibility, and property characteristics, creating a comprehensive dataset for optimization modelling.

2.1.4 Transit Network Data

The integration of Grand River Transit Open Data [7] enabled sophisticated transit accessibility analysis. We examined bus route networks, stop locations, and ION Light Rail Transit infrastructure to develop a multi-modal accessibility scoring system. This integration proved crucial for understanding the relationship between public transit patterns and optimal charging infrastructure placement.

2.2 Population and Electric Vehicle Data

Our demographic analysis synthesized data from Statistics Canada's 2021 Census [12] with electric vehicle ownership patterns from Ontario Data Catalogue's FSA-level database [4]. The census data provided critical information about population distribution, tract boundaries, and housing characteristics across the region. This demographic foundation was enriched by detailed EV registration data, allowing us to map current ownership patterns and project future infrastructure needs.

Chapter 3

Data Processing

The complexity of optimal charging infrastructure placement necessitated a sophisticated data processing methodology. Our approach weighted multiple factors to generate meaningful demand points for the optimization model. Each factor was weighted based on its impact on charging station utilization and long-term viability:

- **EV Ownership Trends (35%):** Derived from FSA-level registration data, received the highest weighting due to their direct correlation with charging demand.
- **Infrastructure Quality (25%):** Infrastructure quality assessment, based on IESO grid capacity data, to ensure feasible implementation.
- **Population Density (20%):** Census tract population distributions from Statistics Canada's Census Profile [12].
- **Regional Transit Accessibility (15%):** Distance to major transit nodes from data provided by Grand River Transit Open Data.
- **Infrastructure Age (5%):** Based on building age data from census. [12]

For demand point generation, we:

1. Created population-weighted centroids for each census tract
2. Applied our multi-factor scoring system
3. Generated optimization model inputs
4. Validated coverage calculations

Chapter 4

Problem Formulation

The optimization of charging infrastructure placement presents a complex multi-objective challenge requiring careful mathematical formulation. We developed a MIP (Mixed-Integer Program) that balances coverage maximization with cost efficiency while adhering to practical infrastructure constraints. This formulation addresses the dual challenges of expanding charging coverage while maintaining cost efficiency through strategic station placement and infrastructure upgrades.

4.1 Data

1. **Sets and Indices:** Our model incorporates existing infrastructure categorization into Level 3 stations (S^3) and Level 2 stations (S^2) eligible for upgrades. The geographical framework encompasses potential locations for new stations (P) and population-based demand points (D), with each demand point carrying a normalized weight (w_j) reflecting its demographic significance. Coverage assessment utilizes sets P_j^2 and P_j^3, representing potential locations capable of serving demand points within Level 2 and Level 3 charging radii, respectively.

   For a detailed listing of sets and indices used in our formulation, see Table 1 in the Appendix.

2. **Coverage parameters** define service areas through radii r^2 and r^3 for Level 2 and Level 3 stations, with minimum coverage requirements α^2 and α^3 ensuring adequate service distribution.

These values, detailed in Table 2 of the Appendix, were established based on standard urban mobility patterns and charging technology specifications from National Resources Canada. [9]

3. **Cost parameters**, outlined in Table 3 of the Appendix, encompass fixed costs c_2^s and c_3^s for new station installations, alongside per-port costs c_2^p and c_3^p. The model incorporates a resale value factor β for equipment recovery during Level 2 to Level 3 upgrades, with all expenditures constrained by the total available budget B. [9]

   These financial parameters were derived from current market data and industry cost analyses. [10]

4. **Infrastructure parameters** define technical specifications, including level-specific minimum charging port requirements p_2 and p_3, and power requirements g_2 and g_3 per charging port. These specifications must respect local grid capacity G_i at each location, as detailed in Table 4 of the Appendix.

   The values align with grid capacity assessments from the IESO Kitchener-Waterloo-Cambridge-Guelph Region Integrated Regional Resource Plan [11].

4.2 Decision Variables

Our model employs binary decision variables to determine infrastructure modifications. Variables x_i^2 and x_i^3 govern new Level 2 and Level 3 station placements, respectively, while u_k determines upgrades of existing Level 2 stations to Level 3 capability. Coverage achievement tracking utilizes variables y_j^2 and y_j^3 to indicate service provision at each demand point for Level 2 and Level 3 charging facilities (stations and/or ports), respectively.

For a comprehensive listing of decision variables and their constraints, refer to Table 5 in the Appendix.

4.3 Constraints

Four essential constraint categories ensure practical feasibility and operational efficiency in our model.

1. The **Station Placement Type constraints** prevent redundant infrastructure by limiting each location to at most one charging station type. See Inequality 4.1.

$$x_i^2 + x_i^3 \leq 1, \quad \forall i \in P \tag{4.1}$$

2. The **Budget constraint** encompasses installation costs for new stations and charging ports, alongside the net cost of upgrades after accounting for equipment resale value. See Inequality 4.3.

$$\begin{aligned} C = &\sum_i (c_2^s \cdot x_i^2 + p_2 \cdot c_2^p \cdot x_i^2) + \\ &\sum_i (c_3^s \cdot x_i^3 + p_3 \cdot c_3^p \cdot x_i^3) + \\ &\sum_{i \in S^2} (u_i (c_3^s + p_3 \cdot c_3^p - \beta \cdot c_2^s - \beta \cdot p_i \cdot c_2^p)) \end{aligned} \tag{4.2}$$

$$C \leq B \tag{4.3}$$

3. **Coverage constraints** ensure adequate service levels across the region. Level 2 coverage requirements consider both new installations and retained ports from upgraded stations, while Level 3 coverage accounts for new installations and upgraded facilities. See Inequalities 4.4 and 4.5.

$$y_j^2 \leq \sum_{i \in P_j^2} x_i^2 + \sum_{i \in S^2} (1 - u_i) \qquad \forall j \in D \tag{4.4}$$

$$y_j^3 \leq \sum_{i \in P_j^3} x_i^3 + \sum_{i \in S^2} u_i \qquad \forall j \in D \tag{4.5}$$

These constraints utilize population-weighted requirements to ensure equitable service distribution. See Inequalities 4.6 and 4.7.

$$\sum_{j \in D} (w_j \cdot y_j^2) \geq \alpha_2 \tag{4.6}$$

$$\sum_{j \in D} (w_j \cdot y_j^3) \geq \alpha_3 \tag{4.7}$$

4. **Grid capacity constraints** ensure that power requirements at each location remain within local electrical infrastructure limitations. These constraints consider both the base load of Level 2 ports and the additional demand from Level 3 charging capabilities, adhering to specifications from Hydro One's regional planning guidelines [5] [6]. See Inequality 4.8.

$$g_2 \cdot p_2 \cdot x_i^2 + g_3 \cdot p_3 \cdot (x_i^3 + u_i) \leq G_i, \quad \forall i \in P \tag{4.8}$$

4.4 Objective Function

Our multi-objective function balances competing priorities through a weighted approach combining maximization of Level 2 service coverage (from 4.6), maximization of Level 3 service coverage (from 4.7), and minimization of total cost (from 4.2).

$$\left\{ \begin{array}{ll} \max & \sum\limits_{j \in D} w_j \cdot y_j^2 \\ \max & \sum\limits_{j \in D} w_j \cdot y_j^3 \\ \min & C \end{array} \right\} \tag{3-OBJ}$$

(3-OBJ) is converted into a weighted linear function. The positive objective weights (λ_1, λ_2, λ_3) reflect strategic priorities while summing to unity. This formulation allows for scenario analysis through weight adjustment while maintaining model linearity.

$$\max \quad \lambda_1 \sum_{j \in D} (w_j \cdot y_j^2) + \lambda_2 \sum_{j \in D} (w_j \cdot y_j^3) - \lambda_3 C \tag{WT-LP}$$

The weights set for our base model are outlined in Table 6 in the Appendix. The polarity between the coverage maximization weights and the cost minimization weights is based on the magnitude of values.

Chapter 5

Results and Analysis

Baseline Infrastructure Analysis

The current charging network in the KWC-CMA region, serving $637,730$ residents [12] across $1,382.1$ km^2 [12], demonstrates significant coverage limitations. Our analysis revealed that the existing 183 charging stations [8] provide inadequate coverage, with Level 2 charging accessible to only 14.57% of the population and Level 3 charging reaching a mere 1.78%. This baseline assessment provided crucial context for optimization targets and infrastructure enhancement strategies.

Optimization Results

The MIP model generated a comprehensive solution addressing current coverage deficiencies while maintaining cost efficiency. The optimized network design incorporates 44 new Level 2 charging stations with single-port configurations in residential areas and community hubs, complemented by 11 new Level 3 fast-charging stations with dual-port configurations along major transportation corridors. Additionally, the model identified four existing Level 2 stations for strategic upgrade to Level 3 capability, augmenting the fast-charging network with eight additional high-capacity ports.

A key feature of our solution is the port retention strategy during upgrades, which ensures continued Level 2 charging availability by maintaining a minimum of one Level 2 port at each upgraded location. This approach balances the benefits of enhanced charging capabilities with the need to serve vehicles that are not compatible with Level 3 charging, while optimizing space utilization and implementation costs.

Implementation of the optimized solution would dramatically improve charging accessibility. Level 2 coverage would increase from 14.57% to 96.14% of the population, while Level 3 coverage would expand from 1.78% to 85.40%. Total charging capacity would increase to 463 Level 2 ports and 96 Level 3 ports, providing robust infrastructure for anticipated EV adoption growth.

Financial Analysis

Economic analysis of the optimized solution revealed a carefully balanced infrastructure investment approach. The total infrastructure investment requirement of $4,100,000 encompasses $1,100,000 for new Level 2 stations, $2,200,000 for new Level 3 stations, and $800,000 for Level 2 to Level 3 upgrades. Equipment resale opportunities of $105,000 reduce the net investment to $3,995,000.

This cost structure, detailed in Figure 3 of the Appendix, demonstrates efficient resource allocation while achieving substantial coverage improvements.

Theoretical Coverage and Sensitivity Analysis

The model revealed theoretical maximum coverage potentials of 95.51% for Level 2 and 98.42% for Level 3 accessibility. Our optimization achieves near-optimal Level 2 coverage at 96.14% and substantial Level 3 coverage at 85.40% while respecting practical constraints.

Sensitivity analysis demonstrated robust solution stability with budget utilization at 99.88% and coverage metrics exceeding minimum targets. See Figure 2 of the Appendix.

Implementation Strategy

The optimization results informed a three-phase implementation strategy designed to systematically enhance the charging network while minimizing operational disruptions.

Phase 1, spanning months 1-4, focuses on strategic Level 2 to Level 3 upgrades, prioritizing grid capacity enhancements and equipment installation at identified high-impact locations. Phase 2 (months 5-8) encompasses new Level 3 station deployments, while Phase 3 (months 9-12) completes the Level 2 network expansion.

This phased approach, illustrated in Figure 4 of the Appendix, ensures coordinated infrastructure development and efficient resource utilization.

Chapter 6

Conclusion

This study presents a comprehensive optimization framework for enhancing electric vehicle charging infrastructure in the Kitchener-Waterloo-Cambridge Census Metropolitan Area. Our Mixed-Integer Program successfully addresses current charging network deficiencies, achieving dramatic improvements in both Level 2 and Level 3 coverage while maintaining cost efficiency. The three-phase implementation strategy, requiring a net investment of $3,995,000, provides a systematic approach to infrastructure enhancement while ensuring efficient resource allocation and minimal service disruptions.

Future research opportunities include the integration of real-time usage data, renewable energy sources, and emerging technologies such as wireless charging. These enhancements would further optimize the network's environmental impact and operational sustainability while preparing for future innovations in EV technology.

The methodology developed in this study provides valuable insights for urban planners and policymakers, offering an adaptable framework that could be extended to other metropolitan areas. As electric vehicle adoption accelerates across Canada, this systematic approach to charging infrastructure optimization will play a crucial role in supporting sustainable urban mobility and achieving national emissions reduction targets.

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APPENDICES

Figures

Figure 1: Data Pipeline

Figure 2: Constraint Sensitivity: Utilization and Slack

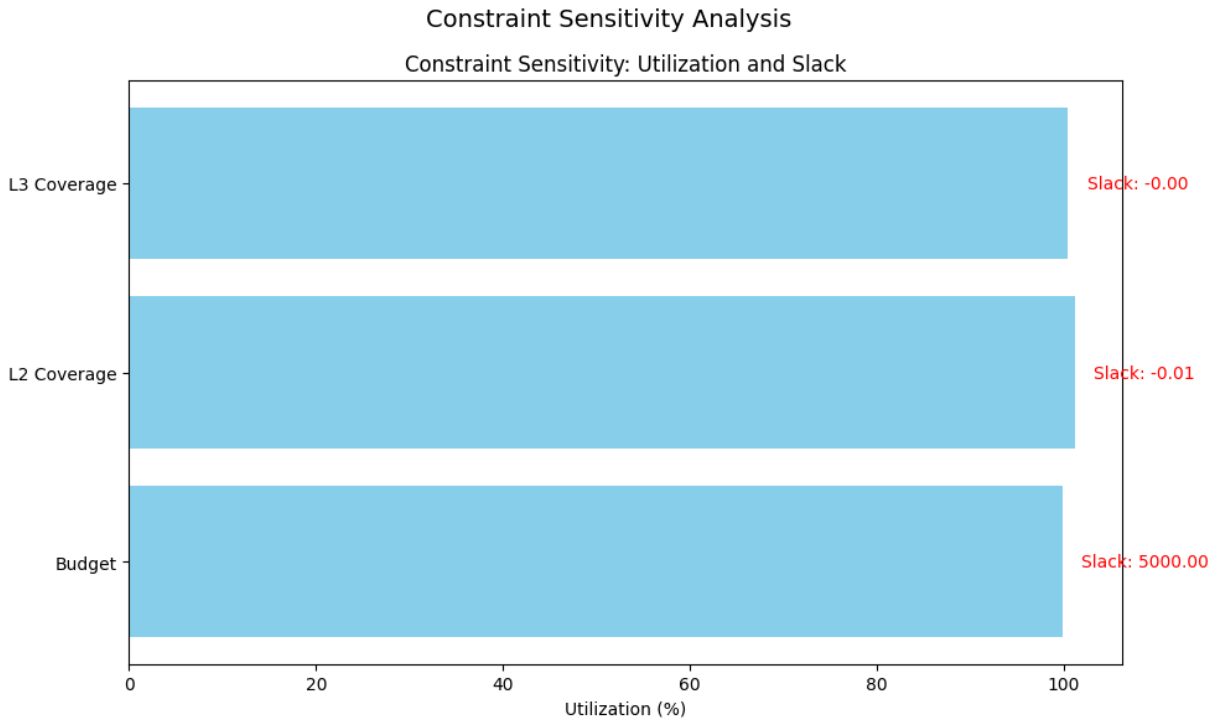


Figure 3: Optimization Results for the model

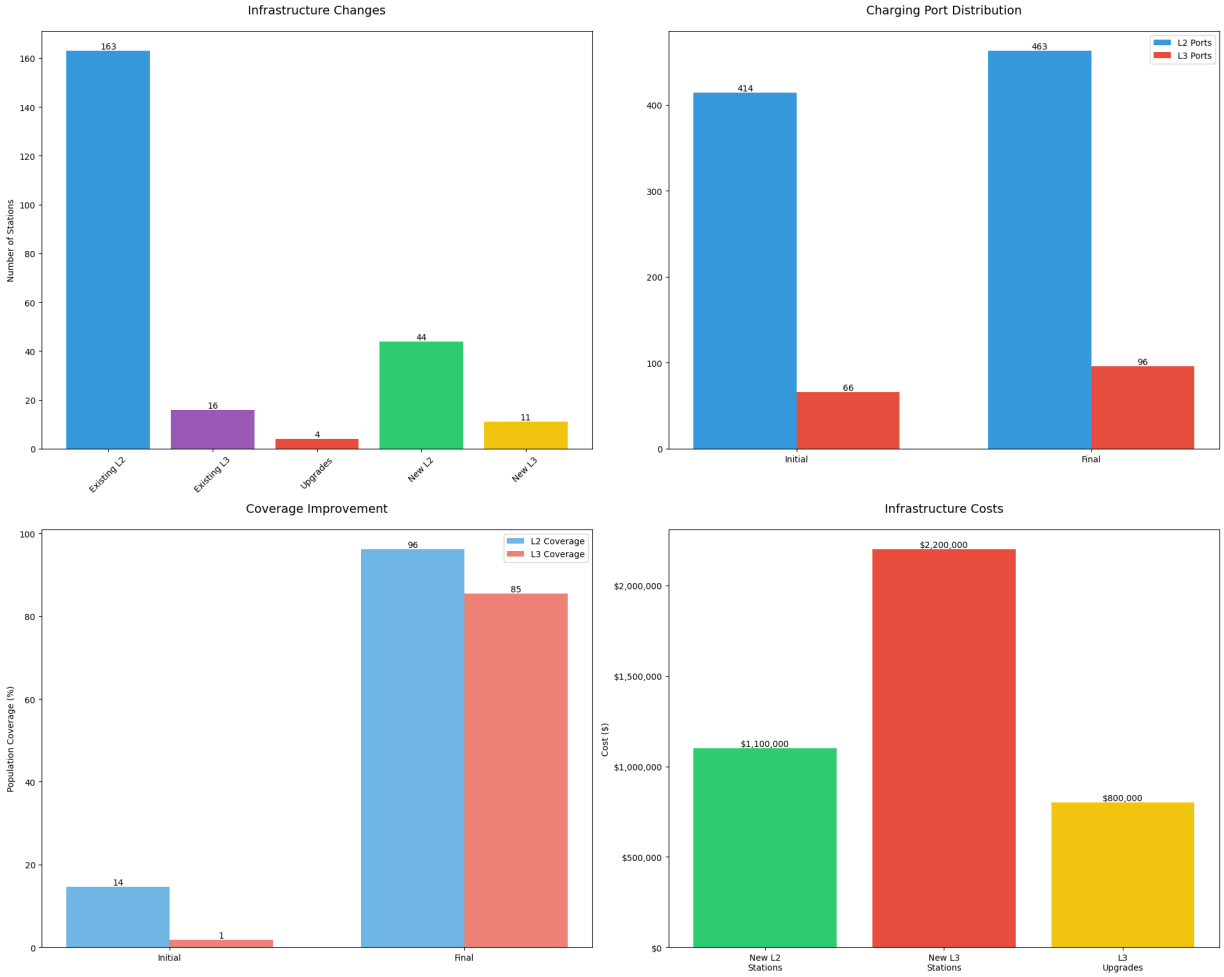
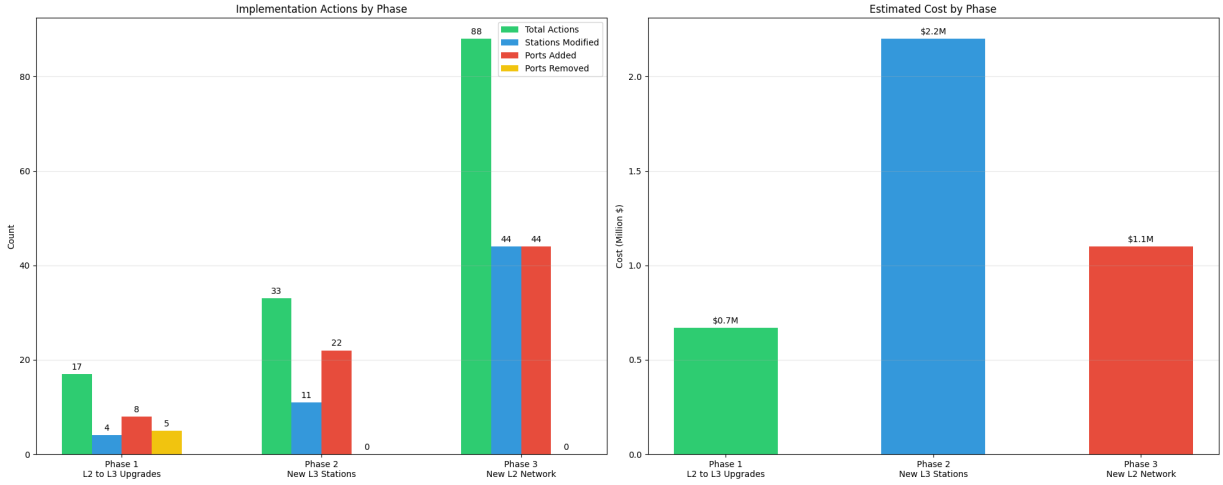


Figure 4: Implementation Plan



Tables

Table 1: Sets and Indices

Set	Indices	Description
S^2	k	Set of existing Level 2 stations (upgrade candidates)
S^3	l	Set of existing Level 3 stations
P	i	Set of potential locations
D	j	Set of demand points
W	w_j	Normalized population weight at demand point $j, \forall j \in D$
P_j^2		Set of potential locations within the Level 2 coverage radius of demand point $j, \forall j \in D$
P_j^3		Set of potential locations within the Level 3 coverage radius of demand point $j, \forall j \in D$

Table 2: Coverage Parameters

Parameter	Base Model Value	Description
r_2	0.50 km	Coverage radius for Level 2 stations
r_3	0.75 km	Coverage radius for Level 3 stations
α_2	95%	Minimum required Level 2 coverage
α_3	85%	Minimum required Level 3 coverage

Table 3: Cost Parameters

Parameter	Base Model Value	Description
c_2^s	\$20,000	Cost of new Level 2 station
c_3^s	\$100,000	Cost of new Level 3 station
c_2^p	\$5,000	Cost per Level 2 charging port
c_3^p	\$50,000	Cost per Level 3 charging port
β	1.0	Resale value factor
B	\$4,000,000	Total available budget

Table 4: Infrastructure Parameters

Parameter	Base Model Value	Description
p_2	1	Minimum Level 2 charging ports per station
p_3	2	Minimum Level 3 charging ports per station
g_2	7.2 kW	Power requirement per Level 2 port
g_3	150 kw	Power requirement per Level 3 port
G_i	500 kw	Grid capacity at site $i, \forall i \in P$

Table 5: Decision Variables

Variable	Indices	Type	Bounds	Description
x_i^2	$i \in P$	Binary	$\{0, 1\}$	Install new Level 2 station at site i
x_i^3	$i \in P$	Binary	$\{0, 1\}$	Install new Level 3 station at site i
u_k	$k \in S^2$	Binary	$\{0, 1\}$	Upgrade existing Level 2 station k to Level 3
y_j^2	$j \in D$	Binary	$\{0, 1\}$	Demand point j is covered by Level 2 charging
y_j^3	$j \in D$	Binary	$\{0, 1\}$	Demand point j is covered by Level 3 charging

Table 6: Base Model (WT-LP) Weight Allocation

Weight	Objective Function Component
66.6666666%	Level 3 Coverage Maximization
33.3333333%	Level 2 Coverage Maximization
0.0000001%	Cost Minimization

Glossary

Grand River Transit Open Data An online resource offering access to transit-related datasets, including routes, schedules, and real-time data for the Grand River area. Useful for developers and analysts working on transportation projects. Access the portal here. 4, 5

Level 1 Level 1 Charging uses standard 120 volts AC power, typically provides 2-5 miles of range per hour. Learn more here 1

Level 2 Level 2 Charging uses 240 volts AC power, typically provides 10-60 miles of range per hour. Learn more here 1–3, 6–12

Level 3 Level 3 Charging, also known as DC Fast Charging, uses 480 volts DC power, typically provides 60-100 miles of range per hour. Learn more here 1–3, 6–12

Ontario Data Catalogue An online platform providing public access to datasets and information from the Government of Ontario. Explore the catalogue here. Ontario Data Catalogue provides an API allowing developers to programmatically search, access, and retrieve datasets from the Ontario Data Catalogue. Learn more it here. 1, 4

Open Data Data made freely available to the public, with no restrictions on use, modification, or sharing. The Ontario Data Catalogue focuses on publishing open data to foster transparency and innovation. 3

OpenChargeMap API OpenChargeMap is a non-commercial, non-profit, EV data service. Their API can be used to retrieve charging location information, user comments, checkins and photos and to submit new comments/checkins. Learn more here. ii, 3

OpenStreetMap A package used to easily download, model, analyze, and visualize street networks and other geo-spatial features. Learn more here. ii, 4

Region of Waterloo's Open Data Portal An online platform provided by the Region of Waterloo offering public access to various datasets, including geographic and infrastructure data. Check it out here. ii, 3

Statistics Canada A platform that provides access to a wide range of Canadian statistical data and metadata. It supports research, analysis, and decision-making in various fields. Explore datasets and tools here. ii, 4, 5

UTM Zone 17N Projection The UTM (Universal Transverse Mercator) is a coordinate system that divides the world into a series of zones for mapping and spatial measurements, specifically referencing Zone 17N in our context. Learn more about it here or from Natural Resources of Canada. 3